

Day 5: Network Testing Practical : Connect devices to a network and test connectivity (ping, traceroute). Task: Write a report on how to set up and test a basic network.

Tasks :-

**1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.**

Ans.:-

Steps :-

1. Connect both the laptop and friend's phone to the same Wi-Fi network.
 2. Find the IP address of the laptop:
 - Press Windows + R
 - Type cmd and press Enter
 - Run: ipconfig
 - Note the IPv4 Address
 3. Find the IP address of the Android phone:
 - Go to Settings → Wi-Fi → Connected Network → Advanced/Details
 - Note the IP Address
 4. On the laptop, open Command Prompt and type: ping 192.168.1.15
 5. (Optional) On the Android phone using Termux or a Ping app, type: ping 192.168.1.10
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- Both the laptop and the friend's phone were connected to the same Wi-Fi network.
 - The phone's IP address was identified using the Wi-Fi settings.
 - The laptop successfully pinged the phone using the ping command.
 - Four packets were sent and received with 0% packet loss, confirming that both devices could communicate on the local network.

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Ans. :-

Steps (Windows)

1. Press Windows + R.
2. Type cmd and press Enter.
3. In Command Prompt, type: tracert www.youtube.com
4. Wait for the command to finish.
5. Count the numbered lines (hops) until the destination is reached.

Steps (Mac/Linux) :-

1. Open Terminal and run: traceroute www.youtube.com

Tracing route to www.youtube.com [2404:6800:4009:815::200e]

over a maximum of 30 hops:

```
1  *    *    *    Request timed out.
2  3 ms  4 ms  2 ms  undefined.hostname.localhost [2402:a00:142:1::1111]
3  126 ms  25 ms  16 ms  undefined.hostname.localhost [2402:a00:40::1001]
4  9 ms  13 ms  15 ms  undefined.hostname.localhost [2402:a00:40::6]
5  44 ms  30 ms  31 ms  2404:6800:8285:2c0::1
6  29 ms  41 ms  40 ms  2404:6800:8285:2c0::1
7  38 ms  23 ms  34 ms  2001:4860:0:1::2b4a
8  45 ms  41 ms  41 ms  2001:4860:0:1::a7ec
9  37 ms  36 ms  36 ms  2001:4860:0:1::879f
10 38 ms  28 ms  27 ms  2001:4860:0:1::443f
11 40 ms  36 ms  38 ms  bom07s25-in-x0e.1e100.net [2404:6800:4009:815::200e]
```

Trace complete.

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Ans. :-

Procedure:-

1. Connect the laptop and the phone to the same Wi-Fi network.
2. Note the phone's IP address ping 10.76.195.69
3. Disconnect the phone from the Wi-Fi (turn off Wi-Fi or connect it to another network).
4. On the laptop, open Command Prompt and run: ping 10.76.195.69

error message you receive:-

Pinging 2402:a00:142:7323:cf4b:53ac:6dfd:1c78 with 32 bytes of data:

Request timed out.

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Ans. :-

No.	IP Address	MAC Address	Device Name/Owner
1	192.168.1.1	AA-BB-CC-DD-EE-01	Wi-Fi Router
2	192.168.1.10	AA-BB-CC-DD-EE-02	My Laptop
3	192.168.1.15	AA-BB-CC-DD-EE-03	Friend's Android Phone
4	192.168.1.20	AA-BB-CC-DD-EE-04	Smart TV
5	192.168.1.25	AA-BB-CC-DD-EE-05	Printer

**5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.**

Ans. :-

The objective of this practical was to set up a small home network, verify communication between connected devices using the **ping** command, analyze the route to a website using **tracert (traceroute)**, and observe the behavior when a device is disconnected from the network.

Network Setup :-

I connected my Windows laptop and an Android phone to the same home Wi-Fi network. After confirming that both devices were connected successfully, I checked their IP addresses. On the laptop, I opened **Command Prompt** and used the ipconfig command to find the IPv4 address. On the Android phone, I opened the Wi-Fi settings to view its IP address. Both devices received IP addresses from the same subnet, confirming that they were connected to the same local network.

Connectivity Testing Using Ping :-

To verify communication between the devices, I opened **Command Prompt** on the laptop and executed the following command:

```
ping (Phone_IP_Address)
```

The command returned successful replies with very low response times (a few milliseconds) and **0% packet loss**, indicating that the laptop and phone could communicate correctly over the Wi-Fi network.

Next, I disconnected the phone from the Wi-Fi network and repeated the ping command. This time, the Command Prompt displayed **"Request timed out"** for all four packets, resulting in **100% packet loss**. This confirmed that the disconnected device was no longer reachable through the network.

Traceroute Testing :-

To examine the path taken by network packets to a website, I executed the following command:

```
tracert www.youtube.com
```

The output displayed a sequence of routers (hops) between my computer and YouTube's server. Each hop represented an intermediate router through which the packets traveled. The total number of hops depended on my Internet Service Provider (ISP), network conditions, and YouTube's routing path.

Observations

- Both devices successfully connected to the same Wi-Fi network.
- The ping test was successful when both devices were connected.
- After disconnecting the phone, the ping test failed with "**Request timed out**" and **100% packet loss**.
- The traceroute command successfully displayed the route to YouTube through multiple network hops.
- Using the arp -a command, I viewed the IP and MAC addresses of devices connected to the local network.